

Sachin Koli

Solution Architect — Cloud Data & AI Platforms

Google Cloud Certified — Professional Data Engineer

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SUMMARY

Google Cloud Certified Professional Data Engineer with **18 years** building enterprise data platforms, now architecting **real-time streaming and batch data infrastructure on Google Cloud** for a UK tier-1 retail bank — Confluent Kafka, Cloud Composer, dbt, BigQuery and Terraform, spanning legacy on-premise and cloud environments.

Five years at that bank across three platforms: digital wallets and Click to Pay, then the migration of its whole card-journey estate to GCP microservices, now the data platform. The self-serve fraud journey I designed won at the **Banking Tech Awards 2024** and the **Card & Payments Awards 2025**.

Across twelve prior years in retail, telecom and banking data: a **Teradata** → **BigQuery** lake migration, and a customer preference API on GCP that **cut \$600,000 a year**. Alongside it I **ship applied-AI tooling in the open** — a terminal coding agent published to npm, and the Rust beneath it: ONNX inference, an ANN index written from scratch, hybrid retrieval.

EXPERIENCE

Tata Consultancy Services

Sep 2021 – Present

Solution Architect · Pune, India

Client: Lloyds Banking Group, United Kingdom — five years, Everyday Banking Platform

GCP Data Platform & Real-Time Analytics — Jul 2025 – Present

- Architecting the bank's move off its **Group Data Warehouse** onto a GCP-native operational data platform against a **Q3-2026 decommissioning date** — authoring the shared **Microbatch Ingestion Service** (GKE, Cloud Storage landing zones, **BigQuery**, **Cloud Composer**, Jenkins CI) that every source system onboards through, with formal **data contracts** for five of them.
- Engineered high-availability, event-driven ingestion on **Confluent Kafka** and **Pub/Sub** with Python, and hybrid batch orchestration spanning **Airflow** and **IBM Tivoli Workload Scheduler** — bridging the legacy on-premise estate to cloud rather than stopping at one or the other.
- Designed BigQuery schemas and modular **dbt** transformation pipelines with delta-load audit tables, enforced data quality and end-to-end lineage; automated testing and monitoring so downstream analytics fail loudly rather than silently.
- Authored the **full architecture-decision baseline** for the bank's digital-behaviour data products — nested schema modelling, partition and cluster strategy per entity, and an identifier strategy that **achieved exact parity with the Adobe Analytics dashboards** while removing the sessionization compute the previous design depended on.
- Standardised provisioning with **Terraform** and designed the platform's FinOps controls; co-owned the **migration recommendation** — ship the MVP on the current GCP foundation to protect the warehouse-exit date, with a pre-planned parallel run onto the strategic one and explicit cutover, rollback and DR criteria.

Card Platform Modernisation on GCP — 2024 – Jul 2025

- Owned the solution design migrating the **entire card-journey estate** — controls, limits, freeze, wallet provisioning, view card and PIN — from on-premise to **GCP microservices behind Apigee X**, sequenced as a **three-tranche** release plan.
- Won approval for the **ID & Auth onboarding pattern** now used across the estate, from a ten-criteria options analysis covering token exchange, proof-of-possession, scaling and traceability.
- Designed **shared scheme-connectivity microservices** for VISA and Fiserv, reused across feature teams on a common certificate-management pattern — every design carried through the bank's architecture, security-assurance and API governance forums.

Digital Wallets, Click to Pay & Card Controls — 2021 – 2024

- Award-winning self-serve fraud journey** — recognised at the Banking Tech Awards 2024 and the Card & Payments Awards 2025 — moving card freeze from an assisted call-centre process to one the customer completes.
- Authored the solution designs putting the bank's four card brands into **Apple Pay, Google Pay and VISA Click to Pay**, including the push-provisioning extension that made wallet-added cards usable for **online** checkout rather than offline only.
- Designed the payment-token security to PCI standards — scheme and wallet **token generation**, PGP key exchange with the wallet provider, HSM-backed keys on **HashiCorp Vault** — plus the lineage, quality and monitoring governance for the wallet data flows.

Sears IT and Management Services India

Feb 2018 – Sep 2021

Technical Specialist → Architect · Pune, India

Client: Sears Holdings Corporation, United States — Targeted Interactions / TI Mart

- Built the **Email Preference Center API** on GCP App Engine, Pub/Sub, DataFlow and BigQuery, serving millions of customers — **reduced annual cost by \$600,000**.

- **Migrated DataMarts from Teradata to BigQuery**, on schedule and without significant incident; designed the shared datamarts (CDW, NSegment, Campaign) on top.
- Built an **API management layer** decoupling applications from cloud complexity across a multicloud estate — reused as the blueprint by other teams — and replaced self-hosted tooling with cloud-native across five stacks: Teradata → BigQuery · Unix → Compute Engine · Control-M → Airflow · shell/ETL → Python · GitLab Enterprise → Cloud Source Repositories.
- Led enterprise migration from **IBM Unica to Adobe Campaign** and Telluride, coordinating vendors (Adobe, BounceX) and internal teams; delivered Tableau and Data Studio reporting and ran continuous GCP cost optimisation.

Cognizant Technology Solutions

Nov 2010 – Feb 2018

Senior Associate – Projects · Pune, India

- Client: Saudi Telecom Company, Saudi Arabia** — STC EDW (CLDM7 / CLDM12) · Onsite IT Delivery Lead · Feb 2015 – Feb 2018
- Led data modelling for the enterprise warehouse — customers, accounts, subscriptions, offers and products across every channel and business unit — extending it by **snowflaking dimensions**.
 - Built an **automation tool** propagating a new key through every dimension table, Teradata utility script (FastLoad, MultiLoad, FastExport, BTEQ) and Ab-Initio object automatically; ran a big-data POC on **Apache Hive and HDFS** for CDR ingestion.
- Client: Barclays, United Kingdom** — Emerging Markets DW & One Africa Program · Team Lead · Nov 2010 – Feb 2015

- Built staging from core banking (FLEX), ATM/POS (SPARROW) and online transaction sources, then the DataMarts above them. **One Africa:** unified emerging-market data across African countries into a single source of truth, letting one DataMart span new countries with no rework.
- Designed the **logical and physical data models**; built reusable automation (TWS object-creation, Teradata code generation, Unix deployment) and archiving that tiered history to **Greenplum**.

Mahindra Satyam

Jan 2008 – Nov 2010

Software Developer – S2 · Hyderabad / Bangalore, India

- Client: Telstra Corporation, Australia** — TR2.1 / TR2.2 EDW Data-In
- Built **DataStage 7.5** jobs, Teradata utility scripts and Perl/Unix tooling for the retail customer warehouse — registrations, demographics, segmentation, billed and unbilled call records — plus **Informatica** mappings migrating Oracle to Teradata.

OPEN SOURCE — APPLIED AI & AGENT TOOLING

Built in my own time and shipped in public — github.com/kolisachint

- **hoocode** — deterministic terminal coding agent, **published to npm**. TypeScript monorepo of four packages: CLI, agent runtime with tool calling, a **unified LLM API across 25+ providers**, differential-rendering TUI. Permission gate on every edit and command, four scoped modes, MCP servers, subagents. ~470 commits since May 2026, extending a fork of the MIT-licensed *pi-mono*.
- **embeddingsearchtools** (Rust) — embedding search engine. MiniLM via **ONNX Runtime**, int8 weights bundled into the binary; exact and **HNSW approximate-nearest-neighbour** indexes written from scratch; **BM25 fusion for hybrid retrieval**; mmap store, library API, CLI, daemon.
- **voicetools** (Rust) — offline speech-to-text: mic capture through voice-activity detection into **Parakeet-TDT on ONNX Runtime** (25 languages, int8), whisper.cpp fallback. **webtools** (Rust) — token-efficient fetch and search for LLM agents; reference-style URL preservation collapses links to single-token markers under a token budget.

TECHNICAL

Cloud	Google Cloud Platform — BigQuery, Cloud Composer, GKE, Pub/Sub, DataFlow, Cloud Storage, Apigee, Compute Engine, Cloud SQL, VPC, IAM · AWS (S3)
Streaming & orchestration	Confluent Kafka, Pub/Sub, Apache Beam · Cloud Composer / Apache Airflow, IBM Tivoli Workload Scheduler, Control-M
Transformation	dbt, Python, SQL, Apache Spark, Informatica, IBM DataStage, Ab-Initio
Architecture & infra	Terraform, CI/CD, containerisation · microservices, API gateways, service discovery, multicloud and hybrid patterns, data governance, cost optimisation
Databases & BI	BigQuery, Teradata, Oracle, Greenplum · Tableau, Looker Studio, Cognos
Domains	Banking, cards & payments · Retail · Telecom

EDUCATION & CERTIFICATIONS

- BE, Computer Science and Engineering** — Walchand College of Engineering, Sangli (Shivaji University), 2003–2007
- **Google Cloud Certified — Professional Data Engineer**
 - Google Cloud Professional Cloud Architect — *in progress*
 - Data Engineering, Big Data and Machine Learning on GCP — Coursera specialisation · Teradata Basics V2R5 · Teradata SQL V2R5 · Introduction to Oracle9i: SQL